A Situation Awareness Perspective on Intelligent Writing Assistants: Augmenting Human-AI Interaction with Eye Tracking Technology

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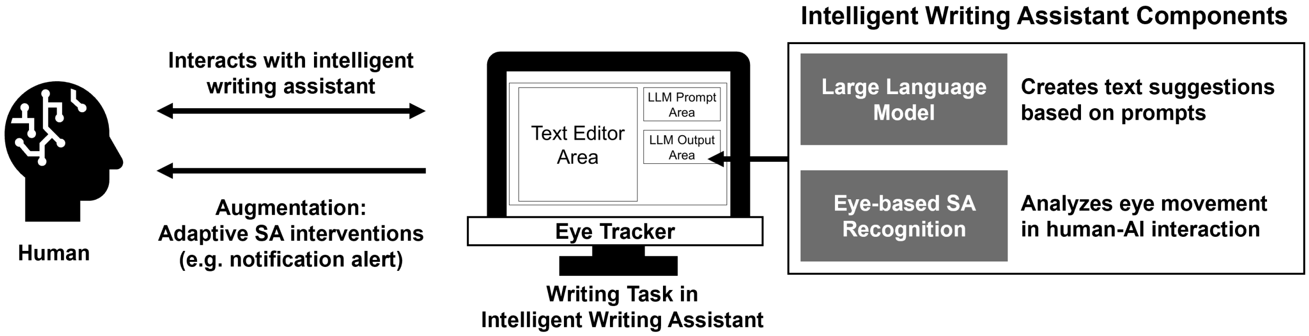

Figure 1: Eye-based Situation Aware Writing Assistant

ABSTRACT

Intelligent writing assistants support the partial automation of the writing process. Existing research has investigated the interaction between humans and automated systems and has identified the maintenance of situation awareness (SA) as a key challenge for humans. Especially in the context of intelligent writing assistants, humans have to maintain SA as they are held responsible for the written text. We build on existing research on automated systems and human-robot/AI collaboration and their interplay with SA theory. In particular, we propose the augmentation of human interaction with intelligent writing assistants through the use of eye tracking technology. Eye tracking technology enables the non-invasive detection of SA based on eye movements. On this basis, writing assistants can be adapted to users' cognitive states such as SA. We argue that for the successful implementation of intelligent writing assistants in the real world, eye-based analysis of SA and augmentation are key.

CCS CONCEPTS

• **Human-centered computing → User centered design**.

KEYWORDS

writing assistant, eye tracking, situation awareness

1 INTRODUCTION

With the rapid development of artificial intelligence (AI) technologies, especially large language models (LLMs) such as ChatGPT, the writing process, originally limited to humans, is now being augmented by machines. In the future, writing assistants will use these AI technologies to automate at least part of the writing process. Writing assistants offer opportunities, but they also pose risks. For example, the pros and cons of this development have been discussed in a recent paper that focuses on the applications of LLMs in education[13]. Kasneci et al. [13] highlighted that ChatGPT could be beneficial for students in the development of their writing, reading and problem solving skills. However, the authors also argue that LLMs could have a negative impact on the critical thinking of both teachers and students, as they may rely too heavily on the output of such models.

Automated generation of human-sounding text may be new, but automation in general, and how people are affected by automated systems, has been studied for decades. In aviation, for example, there has been a great deal of research into how autopilots affect pilot cognition and subsequent behavior [4]. A prominent theory in this area is the Situation Awareness (SA) theory. SA is "conscious knowledge of the immediate environment and the events that are occurring in it. It involves perception of the elements in the environment, comprehension of what they mean and how they relate to one another, and projection of their future states. In ergonomics, for example, it refers to the operator's awareness of the current status and the anticipated future status of a system" [1]. Recently, SA has received increasing attention in human-AI interaction research, such as automated driving [6] and analytical dashboards [15]. SA is a promising theory for understanding and designing

[1] see https://dictionary.apa.org/situation-awareness

human-AI interaction [12] in general and intelligent writing assistants in particular. For example, in the context of writing scientific papers, authors are held responsible for the text generated by AI language tools and included in papers [2]. Therefore, establishing and maintaining SA is critical to detecting errors and hallucinations in the generated text.

SA has traditionally been measured using freezing techniques such as the SA Global Assessment Technique (SAGAT) [9] and the Situation Present Assessment Method (SPAM) [8], or self-report surveys such as the SA Rating Technique (SART) [19]. However, the intrusive and discrete nature of these approaches makes them unsuitable for practical use. Recent research has demonstrated the potential of eye tracking technology to predict SA from eye movements in the specific context of automated driving [22], but also in many other application domains such as aviation, healthcare, and construction [21]. Eye tracking technology is becoming cheaper and more robust, while more devices (e.g., cars, laptops and smartphones with infrared cameras, smart glasses) are incorporating eye tracking capabilities out of the box. Eye tracking technology is therefore a promising solution for assessing SA continuously and in real time. As such, we believe it is a key enabler for situation-aware intelligent writing support.

In this position paper, we present our idea of a situation-aware intelligent writing assistant by augmenting the writing environment with eye tracking technology. In the following, we briefly present the state of the art in eye tracking for writing assistance and SA in human-AI interaction. Based on this, we describe our proposal for situation-aware writing assistance using eye tracking technology. We conclude with a call for interdisciplinary research at the intersection of AI, eye tracking, HCI, and human factors.

2 EYE TRACKING IN SITUATION AWARENESS AND WRITING SUPPORT

According to Endsley [10], SA is established in three hierarchical levels and refers to the state of knowledge. In contrast, situation assessment emphasizes the processes to achieve, acquire, and maintain SA. Level 1 of SA is the perception of relevant elements in the environment. Synthesizing the various perceived elements and understanding them to form a holistic mental model of the environment is considered level 2 understanding. The ability to recognize the status and dynamics of the environment in order to project future actions is achieved at Level 3.

Since SA is grounded in information processing and attention theory, eye tracking is proposed as one of the physiological measures of SA [21]. Eye tracking is a technology capable of capturing the visual attention and cognitive processes of humans [7]. Several studies have investigated different aspects of writing and reading using eye tracking, such as comprehension [3, 17], skimming [2], the reading process during writing [5, 16] and the writing process [11]. Eye tracking technology has also been used to augment the writing process, for example [14] has developed the Eye-Write system to support collaborative writing. In addition, cognitive states such as workload and stress, which are known to affect human SA [10], have been extensively studied using eye tracking technology [1, 20].

[2] https://blog.arxiv.org/2023/01/31/arxiv-announces-new-policy-on-chatgpt-and-similar-tools/

3 OUR PROPOSAL

Figure 1 illustrates our idea of an augmented intelligent writing assistant system that uses a desktop-based eye tracker (e.g., Tobii Pro Nano) for SA detection. The user performs a text writing task in the intelligent writing assistant application by interacting with the LLM. The application consists of a text editor, a prompt area to retrieve text from the LLM, and an output area that provides the generated text from the LLM. Following SA theory, we hypothesize that perception (level 1 of SA) is related to the amount of visual attention spent on the LLM output area. Therefore, level 1 perception should correlate with eye movement metrics such as fixation duration and number of fixations. By using eye tracking technology to detect low SA (missing perception) in real time, the intelligent writing assistant will be able to provide adaptive interventions, e.g. in the form of a notification, to make the user aware of missing SA (level 1). This is an approach to alert the user that the output of the LLM is not well attended and SA needs to be restored.

In order to implement such an SA intelligent writing assistant application, we first plan to collect eye tracking data and SA labels (SART) when using LLMs for text generation in a laboratory experiment. Second, we will analyze the collected data to implement the eye-based recognition of level 1 SA (perception) and build the real-time SA recognition capability to enable the notifications. We plan to compare thresholds for basic eye movement metrics with a machine learning based approach to detect low levels of SA. When low levels of SA are detected, the intelligent writing assistant will provide a notification to increase the user's SA. In addition, we will gain insight into the visual behavior during LLM usage and how users establish and maintain SA, which will enable the design of better user interfaces for such assistants. Finally, we will evaluate the effect of the eye-based SA writing assistant that provides notifications about SA in a laboratory experiment. We hypothesize that such an eye-based SA assistant will improve the quality of writing by making it easier for the user to detect linguistic errors, irrelevant generations, and hallucinations of LLMs. Challenges in developing such an eye-based SA writing assistant will be to identify the right amount and timing of notifications, as this will affect the user experience and usability.

In addition, we will also explore the potential of eye tracking to detect level 2 (comprehension) and level 3 (projection) SA. Continuously calculating and displaying a SA score based on eye tracking data and the attention payed to the LLM output is another promising approach to increasing SA. This will allow for more advanced augmentation capabilities from a SA perspective. In addition, eye tracking can provide insights not only into the user's SA, but also into other cognitive and affective states, such as cognitive load, comprehension, stress, or specific emotions [17, 18], as well as into the user behavior of LLM-based intelligent writing assistants. In the future, we see the potential to improve the user interfaces of LLM-based intelligent writing assistants and to extend the eye-based intelligent writing assistant to detect additional eye-based cognitive and affective states.

4 CONCLUSION

LLMs enable the implementation of intelligent writing assistants in many applications. We argue that there is a need for further research on intelligent writing assistants that leverage the user's perception, understanding and projection of the output of LLMs as writers are held responsible when using LLM output in e.g publications. Therefore, we suggest investigating SA when using intelligent writing assistants and propose to augment intelligent writing assistants with eye tracking technology. The idea is not limited to LLM-based text generation, but can also be applied to other types of writing assistants, such as writing style, grammar, thesaurus assistants, and cognitive and affective user state detection while using writing assistants.

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